

Code No: 125DT**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, November/December - 2017****COMPUTER NETWORKS****(Common to CSE, IT)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- 1.a) Write the advantages of optical fiber over twisted-pair and coaxial cables. [2]
- b) What are the advantages of having layered architecture? [3]
- c) Briefly explain the difference between switch and router. [2]
- d) Sketch the Manchester encoding for the bit stream: 0001110101. [3]
- e) Give the advantages of hierarchical routing. [2]
- f) Differences between CO and CL. [3]
- g) Explain DHCP. [2]
- h) What are the functions of ICMP? [3]
- i) What is the architecture of WWW? [2]
- j) Explain the differences between POP3 and IMAP. [3]

PART - B**(50 Marks)**

- 2.a) Compare and contrast the OSI and TCP/IP reference models.
- b) What are the different types of error detection methods? Explain the CRC error detection technique using generator polynomial x^4+x^3+1 and data 11100011. [5+5]

OR

- 3.a) Discuss about the various transmission media available at the physical layer.
- b) Explain about GBN Sliding Window Protocol. [5+5]

- 4.a) Explain the differences between the switching methods.
- b) Elucidate the CSMA schemes. [5+5]

OR

- 5.a) Illustrate the frame structure of IEEE 802.3.
- b) Give a detail note on the ALOHA protocols. [5+5]

- 6.a) Elucidate Distance Vector Routing Algorithm with example.
- b) Describe the problem and solutions associated with distance vector routing. [5+5]

OR

- 7.a) Explain the general principles of congestion control.
- b) Describe congestion control in datagram subnets. [5+5]

- 8.a) Elucidate the special IP addresses used in internet.
b) Discuss the significance and the operation of NAT. [5+5]

OR

- 9.a) Illustrate the connection establishment and release in transport layer.
b) How crash recovery is managed at the transport layer? [5+5]

- 10.a) Explain Real-time transport protocol.
b) When user clicks a hyperlink, what are the steps that occur between the user's click and the page being displayed? [5+5]

OR

11. Write short notes on the following: [10]
(a) MIME (b) Audio compression (c) DNS (d) Voice over IP.

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